



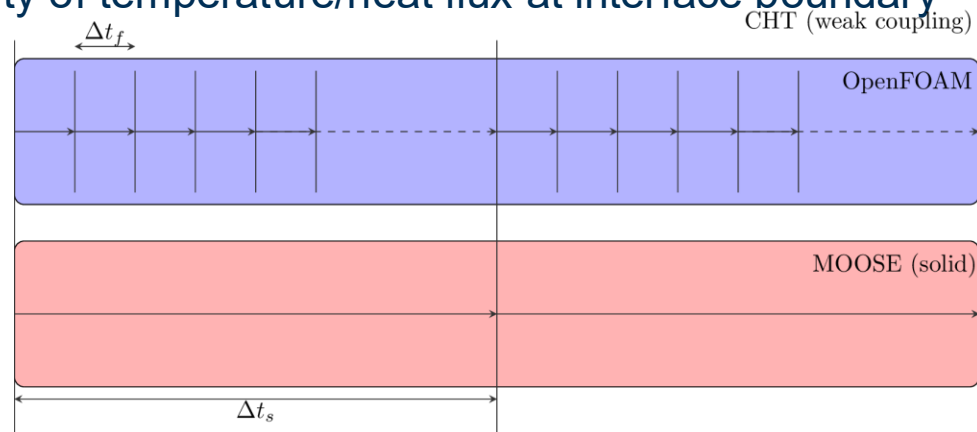
Hippo: conjugate heat transfer

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- Weakly coupled CHT
- Coupling strategies
 - Temperature-forward flux back (TFFB)
 - Flux-forward temperature back (FFTB)
 - Heat transfer coefficient-forward flux back (hFFB)
 - Heat transfer coefficient-forward flux back (hFTB)
- Stability of the coupling strategies
- Exercises
 - Flow over a heated plate
 - TFFB and FFTB
 - Shell and tube heat exchanger
 - hFTB

Weakly-coupled CHT

- Where the fluid and solid thermal fields are solved sequentially.
 - At the end of the step exchange temperature/heat flux.
 - Alternatives include:
 - Solving thermal fields of fluid and solid in the same matrix
 - Use fixed-point iteration to converge boundary temperature/heat flux
- Main advantage is computational cost
 - The fluid field usually requires much smaller time steps than the solid
 - We can take advantage of sub-cycling
- Main disadvantage:
 - Lack of continuity of temperature/heat flux at interface boundary



Coupling strategies for CHT

- Temperature-Forward Flux-Back (TFFB)
 - Temperature sent from fluid to solid
 - Heat flux sent from solid to fluid
- Flux-Forward Temperature Back (FFTB)
 - Heat flux sent from fluid to solid
 - Temperature sent from solid to fluid
- Heat-transfer coefficient Forward Temperature Back (hFTB)
 - HTC and bulk temperature sent from fluid to solid
 - Heat flux sent from solid to fluid
- Heat-transfer coefficient Forward Flux Back (hFFB)
 - HTC and bulk temperature sent from fluid to solid
 - Temperature sent from solid to fluid

Stability of CHT coupling strategies

- The stability of the coupling depends on the conditions on each side of the interface.

- Verstraete and Scholl (2016) highlighted the importance of the Biot number

$$Bi = h L_s / k_s$$

- FFTB and hFTB are stable for $Bi < 1$ and TFFB and hFFB are stable for $Bi > 1$, although with many caveats
 - The analysis doesn't account for transient effects which at some point I want to consider
 - Density and heat capacity of the fluid and solid are also important for stability
- The basic instability mechanism is due to growing heat flux oscillations on the interface
 - Large temperature changes on a time step leads to growing reversals of heat flux

TFFB in Hippo

- Some key parts from theflow over heated plate TFFB example
 - AuxKernel to calculate heat flux on solid-side
 - FoamDiffusionFluxBC imposes it
 - Temperature copied with FoamVariableField
 - MatchedValueBC imposes it

```
[AuxVariables]
[fluid_wall_temperature]
  family = LAGRANGE
  order = FIRST
  initial_condition = 0
[]
[wall_heat_flux]
  family = MONOMIAL
  order = CONSTANT
  initial_condition = 0
[]
[]

[AuxKernels]
[heat_flux_aux]
  type = DiffusionFluxAux
  variable = wall_heat_flux
  diffusion_variable = T
  boundary = 'solid_top'
  diffusivity = thermal_conductivity
  component = normal
[]
[]

[BCs]
[fixed_temp]
  type = DirichletBC
  variable = T
  boundary = solid_bottom
  value = 310
[]

[fluid_interface]
  type = MatchedValueBC
  variable = T
  boundary = solid_top
  v = fluid_wall_temperature
[]
[]
```

```
[Transfers]
[wall_temperature_from_fluid]
  type = MultiAppGeometricInterpolationTransfer
  source_variable = fluid_wall_temp
  from_multi_app = hippo
  variable = fluid_wall_temperature
  execute_on = same_as_multiapp
[]

[heat_flux_to_fluid]
  type = MultiAppGeometricInterpolationTransfer
  source_variable = wall_heat_flux
  to_multi_app = hippo
  variable = solid_heat_flux
  execute_on = same_as_multiapp
[]
[]

[FoamBCs]
[solid_heat_flux]
  type = FoamDiffusionFluxBC
  foam_variable = T
  initial_condition = 0
[]
[]

[FoamVariables]
[fluid_wall_temp]
  type = FoamVariableField
  foam_variable = T
  initial_condition = 300
[]
[]
```

FFTB in Hippo

- Some key parts from theflowover heated plate FFTB example
 - ProjectionAux to create constant monomial
 - FoamFixedValueBC imposes it
 - Heat flux calculated with OpenFOAM function object
 - CoupledVarNeumannBC imposes it

```
[AuxVariables]
[solid_wall_temperature]
  family = LAGRANGE
  order = FIRST
  initial_condition = 300
[]
[fluid_heat_flux]
  family = MONOMIAL
  order = CONSTANT
  initial_condition = 0
[]
[]

[AuxKernels]
[projection_aux]
  type = ProjectionAux
  variable = solid_wall_temperature
  v = T
  boundary = 'solid_top'
[]
[]

[BCs]
[fixed_temp]
  type = DirichletBC
  variable = T
  boundary = solid_bottom
  value = 310
[]
[fluid_interface]
  type = CoupledVarNeumannBC
  variable = T
  boundary = solid_top
  v = fluid_heat_flux
  scale_factor = -1.
[]
[]
```

```
[MultiApps]
[hippo]
  type = TransientMultiApp
  app_type = hippoApp
  execute_on = timestep_begin
  input_files = 'fluid.i'
[]

[Transfers]
[heat_flux_from_fluid]
  type = MultiAppGeometricInterpolationTransfer
  source_variable = fluid_heat_flux
  from_multi_app = hippo
  variable = fluid_heat_flux
  execute_on = same_as_multiapp
[]
[temp_to_fluid]
  type = MultiAppGeometricInterpolationTransfer
  source_variable = solid_wall_temperature
  to_multi_app = hippo
  variable = solid_wall_temp
  execute_on = same_as_multiapp
[]
[]

[FoamBCs]
[solid_wall_temp]
  type = FoamFixedValueBC
  foam_variable = T
  initial_condition = 300
[]

[FoamVariables]
[fluid_heat_flux]
  type = FoamFunctionObject
  foam_variable = "wallHeatFlux"
  initial_condition = 0
[]
[]
```

hFTB in Hippo

- Why might we choose to use hFTB and not FFTB
 - Verstraete and Scholl (2016) suggests stability might be similar
- In some case, we only care about the steady state solution
 - At each coupling step, we can do a steady solve – Poisson equation
 - Generally, much faster convergence
 - However, the Laplace operator is singular with all-Neumann BCs
 - Frequently happens using FFTB and *usually* can't be solved
 - Therefore, it must be run transient.
 - hFTB can be run steady-state
- To use the HTC, we use a special FoamVariable called FoamHeatTransferCoeff
 - It also requires a method of calculating the bulk temperature.
 - In Hippo, these are implemented as UserObjects
 - AdjacentCellBulkTemperature
 - FixedBulkTemperature

hFTB in Hippo

- Note user object and FoamVariable
- CoupledConvectiveHeatFluxBC used to impose the HTC boundary condition
- Bulk temperature UO and the HTC needs to be transferred

```
[AuxVariables]
  [inner_htc]
    family = MONOMIAL
    order = CONSTANT
    initial_condition = 0
  []
  [bulk_t_inner]
    family = MONOMIAL
    order = CONSTANT
  []
```

```
[FoamVariables]
  [fluid_htc]
    type = FoamHeatTransferCoeff
    bulk_temperature_uo = bulk_t
    boundary = interface
  []

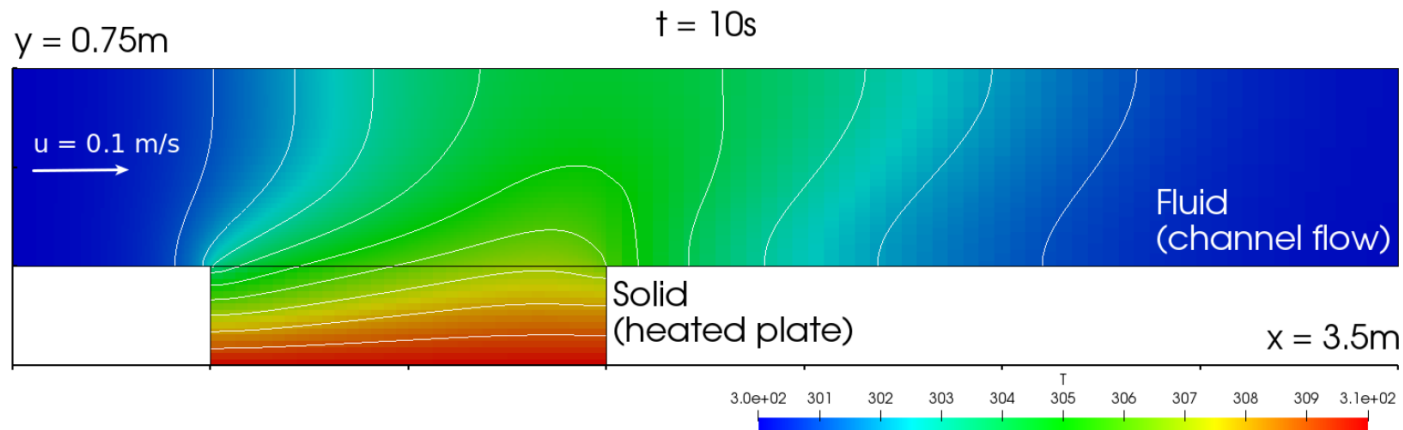
[UserObjects]
  [bulk_t]
    type = AdjacentCellBulkTemperature
    boundary = interface
    execute_on = 'initial timestep_end'
  []
[]
```

```
[inner]
  type = CoupledConvectiveHeatFluxBC
  variable = T
  boundary = inner
  T_infinity = bulk_t_inner
  htc = inner_htc
[]
```

```
[htc_from_outer]
  type = MultiAppGeneralFieldNearestLocationTransfer
  source_variable = fluid_htc
  from_multi_app = outer
  variable = outer_htc
  execute_on = same_as_multiapp
  to_boundaries = 'outer'
  search_value_conflicts=false
[]
[bulk_from_inner]
  type = MultiAppGeneralFieldUserObjectTransfer
  source_user_object = bulk_t
  from_multi_app = inner
  variable = bulk_t_inner
  bbox_factor = 2.
  to_boundaries = 'inner'
  greedy_search = false
[]
#
```

Example 1: Flow over a heated plate

- Derives from preCICE example
- Transient heat conduction in the heated plate modelled using MOOSE
 - Using TFFB and FFTB
- For FFTB, we will consider different physical properties highlighting the stability of the CHT schemes



Example 2: Shell and tube heat exchanger

- Also derives from a preCICE tutorial
- Two OpenFOAM simulations
 - Tube (inner)
 - Shell (outer)
- MOOSE used for solid heat conduction
 - Replaces CalculiX in the tutorial
 - Both use the same meshes as the preCICE example
- Uses hFTB with solid run steady-state

